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Problem
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You are given a directed flow network consisting of V vertices and E directed edges. Each edge has a capacity that represents the maximum amount of flow that can pass through it.

The source vertex is 0 and the sink vertex is V-1.

Your task is to find the maximum possible flow from the source to the sink using the Ford—Fulkerson algorithm.

**Input Format**

The first line contains two space-separated integers: V — number of vertices E — number of directed edges Each of the next E lines contains three space-separated integers: u — starting vertex v — ending vertex capacity — capacity of the edge

**Constraints**

2 ≤ V ≤ 100 1 ≤ E ≤ 1000 0 ≤ u, v < V u ≠ v 1 ≤ capacity ≤ $10^6$

**Output Format**

Print a single integer representing the maximum flow from vertex 0 to vertex V-1.

**Sample Input 0**

```
6 9
0 1 16
0 2 13
1 2 10
2 1 4
1 3 12
3 2 9
2 4 14
4 3 7
3 5 20
```

**Sample Output 0**

```
19
```

```python
from collections import deque
def ford_fulkerson(g, n):
    flow = 0
    while True:
        parent = [-1] * n
        parent[0] = 0
        q = deque([0])
        while q:
            u = q.popleft()

            for v in range(n):
                if parent[v] == -1 and g[u][v]
                    parent[v] = u
                    q.append(v)

        if parent[n - 1] == -1:
            break
        f = float('inf')
        v = n - 1
        while v != 0:
            u = parent[v]
            f = min(f, g[u][v])
            v = u
        v = n - 1

        while v != 0:
            u = parent[v]
```

Line: 46 Col: 1

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Test against custom input

## Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Input (stdin) Download

```
6 9
0 1 16
0 2 13
1 2 10
2 1 4
1 3 12
3 2 9
2 4 14
4 3 7
3 5 20
```